

SQL DATABASE FUNDAMENTALS (345)

STATE – 2019

TOTAL POINTS _____ ***(500 points)***

Failure to adhere to any of the following rules will result in disqualification:

- 1. Contestant must hand in this test booklet and all printouts. Failure to do so will result in disqualification.**
- 2. No equipment, supplies, or materials other than those specified for this event are allowed in the testing area. No previous BPA tests and/or sample tests or facsimile (handwritten, photocopied, or keyed) are allowed in the testing area.**
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No more than sixty (60) minutes testing time

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MULTIPLE CHOICE

Identify the letter of the choice that best completes the statement or answers the question. Mark "A" if the statement is true. Mark "B" if the statement is false.

1. The following example utilizes proper syntax to grant permission to the database user *Issac* to alter a database.

```
USE master;  
GO  
GRANT ALTER DATABASE TO [Issac];  
GO
```

- a. True
 - b. False
2. All mathematical functions are deterministic.
- a. True
 - b. False
3. Which of the following apply to the SUM aggregate function in SQL Server?
- a. Compute the sum of all values within the function including Null values
 - b. DISTINCT is not a valid argument
 - c. Returns data type int or big money
 - d. Computes the sum of all non-null values within an expression
4. Which of the following is *not* true about the AVG function in SQL Server?
- a. Unique Values are not considered within the DISTINCT argument
 - b. The Group By argument computes the AVG function, based on the group and not the whole table
 - c. The over argument divides the result set based upon the FROM clause
 - d. ALL is the default argument
5. In SQL Server, which of the following is a *valid* ALTER statement?
- a. ALTER INDEX
 - b. ALTER CONSTRAINT
 - c. ALTER TABLE
 - d. ALTER ROLE
6. Which of the following statements is used to select the existing database *TEST* in SQL schema?
- a. USE TEST
 - b. ALTER TEST
 - c. OPEN TEST
 - d. SELECT TEST

7. Which of the following are SQL conjunctive operators?
- a. OR, AND
 - b. NOT, IS NULL
 - c. AND, BETWEEN
 - d. ALL, ANY
8. The formal term for saving changes invoked by a transaction to the database is a(n) _____.
- a. alter
 - b. rollback
 - c. commit
 - d. execute

9. The following ALTER TABLE statement _____.

```
ALTER TABLE TestTaking ADD Column  
Testscore Int,  
Testdate datetime NOT NULL DEFAULT GETDATE(),
```

- a. alters the table
 - b. creates a column with the data type int
 - c. creates a column to store test date based upon when the test was taken
 - d. all the above
10. A multivalued dependency occurs when _____.
- a. dependency of an attribute on a set of attributes
 - b. dependency formed by two functional dependencies
 - c. attributes dependent upon each other
 - d. more than one independent multivalued attributes within a table
11. Non-Clustered indexes sort and store data rows within the table based upon their key values.
- a. True
 - b. False
12. Clustered and Non-Clustered indexes can be unique so that no two rows can have the same values within the index key.
- a. True
 - b. False
13. A table that doesn't have a clustered index is referred to as a _____.
- a. Clustered table
 - b. User-Defined table
 - c. Heap table
 - d. Non-Clustered table

14. In SQL, indexes that have one or more columns are referred to as which of the following?
- Unique index
 - Composite index
 - Clustered index
 - Table index
15. Besides Clustered and Non-Clustered indexes, other types of indexes include composite, unique, and covering.
- True
 - False
16. The acronym ACID stands for the following?
- Analysis, Console, Intrusion, Durability
 - Acidification, Chemistry, Information, Database
 - Atomicity, Consistency, Isolation, Durability
 - Advanced, Computing, Information, Database
17. In a relational database, a consistent transaction is _____.
- a series of actions that needs to be completed in order for the transaction to go through
 - an operation that delete all tables
 - one that creates the structure of a certain database table
 - one where all data needs to be valid according to all rules including constraints, cascades, and triggers
18. Primary key constraints ensure _____.
- domain integrity
 - referential integrity
 - user-defined integrity
 - entity integrity
19. The SQL keyword that is used to select records from two tables that have matching values is _____.
- Self-Join
 - Inner Join
 - Outer Join
 - Full Join
20. The SQL operator that is used to combine the result sets of two or more SELECT statements is _____.
- Union
 - Join
 - Having
 - Exists

21. Which of the following is *not* a subset of DQL?
- a. HTTP
 - b. PQL
 - c. OQL
 - d. MDX
22. The SQL function that is used to bring back a certain part of a date as an integer value is:
- a. DATEADD
 - b. GETUTCDATE
 - c. DATEPART
 - d. TIMESTAMP
23. Consider the following SQL statement
- ```
CREATE VIEW Customers
```
- The SQL statement \_\_\_\_\_.
- a. generates an error
  - b. creates a view named Customers
  - c. creates a table named Customers
  - d. does nothing
24. Which of the following is *not* an SQL trigger type?
- a. DML
  - b. DDL
  - c. CLR
  - d. DAL
25. The following is the correct order of processing within a typical sql statement: SELECT, WHERE, GROUP BY, HAVING
- a. True
  - b. False
26. Which of the following statements regarding views is *false*?
- a. Can be used for hiding complexity
  - b. DML queries can't be performed, if there are more than one table within the view
  - c. When you drop the base table, the view becomes inactive
  - d. All of the above statements are true
27. Which of the following command(s) is/are related to transaction control in SQL?
- a. SAVEPOINT
  - b. ROLLBACK
  - c. COMMIT
  - d. All of the above

28. A subquery in an SQL SELECT statement is enclosed in \_\_\_\_\_.  
a. Braces { }  
b. Brackets [ ]  
c. Parenthesis ( )  
d. None of the above
29. The DMLs that require users to specify what data is needed and how to get it, are called:  
a. Non-procedural DMLs  
b. Procedural DMLs  
c. DDL DMLs  
d. None of the above
30. Which of the following statement is *not* true about embedded SQL?  
a. In embed SQL, all query processing is performed by the database system  
b. Embedded SQL expressions for database modification always returns a result  
c. An embedded SQL program must be processed by a special processor prior to compilation  
d. The exact syntax for embedded SQL request depends on the language in which SQL is embedded
31. Which of the following SQL commands *will* include all records from one table and match it with a second table?  
a. LEFT OUTER JOIN  
b. RIGHT OUTER JOIN  
c. Both A and B  
d. Neither A or B
32. SQL Server recognizes which of the following classes of constraints?  
a. Index  
b. Check  
c. Not Null  
d. All of the above
33. The SCHEMA\_ID in SQL Server \_\_\_\_\_.  
a. returns the schema ID associated with the schema name  
b. returns the schema ID associated with the schema status  
c. returns the schema ID associated with the schema number  
d. None of the above
34. What effect will a ROLLBACK statement have after a COMMIT has been issued?  
a. No effect, COMMIT ends the transaction.  
b. Issuing a prompt to make sure you want to commit the rollback.  
c. All statements within the transaction are rolled back.  
d. All the above

35. The following statement creates a copy of the data from one table, and inserts it within another table, so that the data types in both the source and target tables match.
- INSERT
  - INSERT INTO SELECT
  - SELECT INSERT
  - None of the above
36. The T-SQL REVOKE clause is used to “REVOKE” <some permission> ON <some object> FROM <some user, login, or group>.
- True
  - False
37. Which of the following statements is *true* about the HAVING clause?
- The HAVING keyword collects data across multiple records and groups them by one or more columns.
  - The HAVING keyword removes duplications from the results.
  - The HAVING keyword is used in conjunction with the GROUP BY keyword to filter out certain rows.
  - None of the above
38. When three or more AND and OR operators are combined, it is easier to use the following SQL keyword(s) \_\_\_\_\_.
- IN Only
  - NOT LIKE Only
  - Like Only
  - Both IN and NOT IN
39. The following SQL is which type of join: SELECT CUSTOMER\_T.CUSTOMER\_ID, ORDER\_T.CUSTOMER\_ID, NAME, ORDER\_ID FROM CUSTOMER\_T, ORDER\_T WHERE CUSTOMER\_T.CUSTOMER\_ID = ORDER\_T.CUSTOMER\_ID?
- Left Join
  - Inner Join
  - Outer Join
  - Equi-join
40. Which of the following statements is *true* concerning subqueries?
- Do not start with the word SELECT
  - Involves the use of an inner and outer query
  - Cannot return the same result as a query that is not a subquery
  - All of the above
41. What type of lock will deny users any access to a table?
- Implicit
  - Shared
  - Exclusive
  - Explicit

Consider the following code for the questions 42-45.

- a) IF EXISTS (SELECT \* FROM sys.tables WHERE name = 'dbo.TestTable')
- b) DROP TABLE dbo.TestTable
- c) GO
- d) CREATE TABLE dbo.ActualTable
- e) (
  - f) col\_1 CHAR(1),
  - g) col\_2 DATE,
  - h) col\_3 TIME,
  - i) col\_4 VARCHAR (10)
- j) );
- k) GO
- l) INSERT INTO dbo.ActualTable (col\_1, col\_2, col\_3, col\_4)
- m) VALUES ('D', '2018-05-07', '15:00', 'Contest')
- n) GO
- o) SELECT \* FROM dbo.ActualTable

42. The INSERT in line l, performs the function of insert values into the columns that are within the table.
- a. True
  - b. False
43. What statement is *true about* syntax VARCHAR, in line i?
- a. Holds a string with a maximum length of 10 characters
  - b. Variable width Unicode string data type
  - c. The ability to hold a variable amount of characters
  - d. None of the above
44. The code in line a, checks to see if a certain table doesn't exist within the database.
- a. True
  - b. False
45. The 'dbo' in a line refers to \_\_\_\_.
- a. User Name
  - b. Object Name
  - c. Table Schema
  - d. None of the above
46. What will be the result of: Select e.Department, sum(h.Normalhours) from Employee as e inner join HoursVolunteered as h on e.EmpID = h.EmpID where e.Department like 'IT' Group by e.Department \_\_\_\_.
- a. shows a total of normal hours volunteered by all departments
  - b. shows average of normal hours worked by the IT department
  - c. shows a total of normal hours volunteered by the IT department
  - d. displays a list of normal hours volunteered by the IT department

47. Primary key values must be unique.
- a. True
  - b. False
48. Which of the following performs the ability to insert a data file within a database table as a user-specified format within SQL?
- a. INSERT
  - b. BULK INSERT
  - c. CREATE
  - d. MERGE
49. Who is considered as the “father” of the relational model for database management?
- a. Steve Jobs
  - b. Bill Gates
  - c. Brian Krzanich
  - d. Edgar Codd
50. The SQL ORDER BY clause \_\_\_\_\_.
- a. sorts records among a column either in ascending or descending order
  - b. WHERE conditions ASC or DESC are optional within an ORDER BY clause
  - c. if no ASC or DESC modifier is not provided, the results will still be sorted in ascending order
  - d. All the above